

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester B.Tech (CBCS) Degree Examinations
 December 2023
EE1002-1 – BASIC ELECTRICAL AND ELECTRONICS ENGINEERING
 (For CV, ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- In the case of nodal analysis, the equations at each node is written by applying _____
 A) KVL B) KCL
 C) Both KVL and KCL D) None of these
- SI unit of electric charge is _____
 A) ampere B) volt
 C) coulomb D) joule
- 50Hz alternating voltage has an angular velocity of _____ rad/sec.
 A) 100π B) 50π
 C) $2\pi/50$ D) None of these
- Power factor of a pure resistive circuit is _____
 A) Zero B) Unity
 C) 0.5 D) None of these
- The unit of capacitive reactance is _____
 A) ohm B) farad
 C) mho D) None of these
- The number of parallel paths in 8 pole wave wound DC motor is _____
 A) 2 B) 8
 C) 4 D) 16
- Two resistors of resistances $R_1 \Omega$ and $R_2 \Omega$ are connected in series in a network. The equivalent resistance R_{eq} is given by
 A) $\frac{R_1 R_2}{R_1 + R_2}$ B) $\frac{1}{R_1} + \frac{1}{R_2}$
 C) $R_1 + R_2$ D) $R_1 - R_2$
- A transformer has 100 primary turns and 50 secondary turns if the primary voltage is 200V then the secondary voltage is _____
 A) 100 V B) 400 V
 C) 200 V D) $\frac{1}{400}$ V
- For current working of an NPN bipolar junction transistor, the different electrodes should have the following polarities with respect to emitter.
 A) collector +ve, base +ve B) collector -ve, base -ve
 C) collector +ve, base -ve D) collector -ve, base +ve
- The value of total collector current in a CB circuit is
 A) $I_C = \alpha I_E$ B) $I_C = \alpha I_E + I_{CO}$
 C) $I_C = \alpha I_E - I_{CO}$ D) $I_E = \alpha I_C$

11. The following relationships between α and β are correct EXCEPT
- A) $\beta = \frac{\alpha}{1 - \alpha}$ B) $1 - \alpha = \frac{1}{1 + \beta}$
- C) $\alpha = \frac{\beta}{1 + \beta}$ D) $\alpha = \frac{\beta}{1 - \beta}$
12. Zener diodes are used primarily as
- A) rectifiers B) voltage regulators
- C) oscillators D) Amplifiers
13. A bipolar junction transistor is a
- A) Current controlled device B) Voltage controlled device
- C) Resistance controlled device D) None of these
14. In the breakdown region, Zener diode behaves like a source of
- A) Constant current B) Constant power
- C) Constant resistance D) Constant voltage
15. When a BJT operates in cut-off
- A) $V_{CE} = 0$ B) $V_{CE} = V_{CC}$
- C) V_{CE} has the negative value D) I_C is maximum
16. When a BJT is in saturation
- A) $I_C = 0$ B) I_B controls I_C
- C) $V_{CE} = 0$ D) V_{CE} has the positive value
17. A simple application of an Op-Amp that can be used to generate a gain of unity is
- A) Differentiator B) Integrator
- C) Comparator D) Voltage follower
18. The pin number 7 in IC 555 timer is assigned as
- A) Threshold pin B) output pin
- C) Discharge pin D) None of these
19. An ideal Op-Amp has following characteristics
- A) $R_i = \infty, A_v = \infty, R_o = 0$ B) $R_i = 0, A_v = \infty, R_o = 0$
- C) $R_i = \infty, A_v = 0, R_o = \infty$ D) $R_i = 0, A_v = \infty, R_o = \infty$
20. A non-inverting closed loop Op Amp circuit generally has a gain factor
- A) > 1 B) < 1
- C) $= 1$ D) None of these

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) With neat diagram, explain the constructional features of three-phase induction motor.	08	L*1	2	1,2
	b) Derive the equation for average value of sinusoidally varying quantity.	08	L3	1	1,2
2.	a) With neat diagram, explain the working principle of transformer	06	L1	2	1,2
	b) An AC Circuit consists of a pure resistance of 50Ω and is connected across an AC supply of 230 V, 50Hz. Calculate current, power consumed, equation for voltage and current.	06		1	1,2
	c) A 250 kVA, 11 000 V/400 V, 50 Hz single-phase transformer has 80 turns on the secondary. Calculate:				
	i) the approximate values of the primary and secondary currents;				
	ii) the approximate number of primary turns;				
	iii) the maximum value of the flux.	04	L1	2	1,2
3.	a) Apply the mesh analysis method to the network of Figure Q3(a), find Power absorbed by 4Ω resistor.				

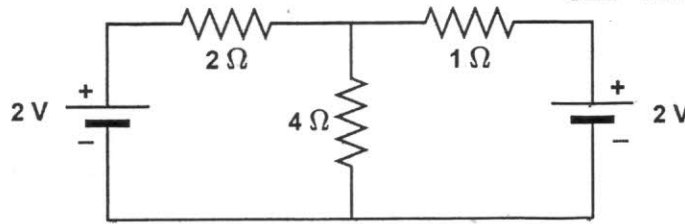


Figure Q3(a)

- b) Derive the equation for torque developed in the armature of a DC motor.
 c) With neat diagram, explain the construction of core type single phase transformer.

04	L3	1	1,2
06	L2	2	1,2
06	L1	2	1,2

Unit – II

4. a) With neat diagram and waveforms, explain the working of single-phase half wave rectifier.
 b) Explain the application of BJT as a switch.
5. a) What is a Zener diode and write a note on I-V characteristics of Zener diode.
 b) Derive the equation of emitter current I_E for common emitter configuration of BJT.
6. a) With neat diagram and waveforms, explain the working of single-phase full wave rectifier
 b) Explain application of BJT circuit to switch ON/OFF an LED

08	L2	3	1,2
08	L3	4	1,2
08	L1	3	1,2
08	L3	4	1,2
08	L2	3	1,2
08	L3	4	1,2

Unit – III

7. a) Explain how an Op-Amp can be used as a differentiator.
 b) Explain the operation of Op-Amp as a non-inverting amplifier.
8. a) Explain the Astable mode of 555 timer.
 b) For the circuit of inverting amplifier shown in Figure 8(b), calculate the following:
 i) Closed loop gain A_f
 ii) Output Voltage V_o

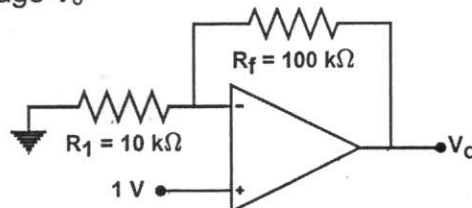


Figure 8(b)

- c) For the circuit of inverting amplifier shown in Figure 8(c), calculate the following:
 i) Closed loop gain A_f
 ii) Output Voltage V_o
 iii) Input Current I_1

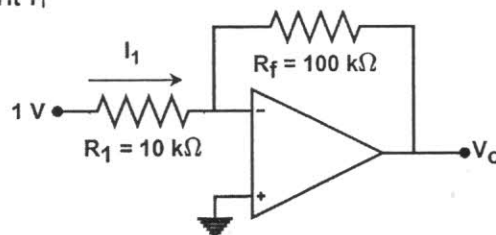


Figure 8(c)

04	L2	5	1,2
04	L2	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
